

Advait Deshmukh

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Boulder, CO, USA

RESEARCH INTERESTS

Natural Language Processing, Narrative Understanding, Computational Discourse, Interpretability

EDUCATION

Doctor of Philosophy, Computer Science

University of Colorado Boulder

Advised by Dr. Maria Antoniak

Jan 2026 – Dec 2030

Boulder, CO

Master of Science, Computer Science

University of Colorado Boulder

Advised by Dr. Maria Leonor Pacheco

Aug 2023 – Dec 2025

Boulder, CO

Bachelor of Engineering, Computer Science

Birla Institute of Technology and Science, Pilani

Aug 2017 – Aug 2021

Pilani, India

PUBLICATIONS

C=CONFERENCE, S=IN SUBMISSION

- [C.1] Advait Deshmukh, Nora Benedict, Melanie Walsh, and Maria Antoniak. 2026. The Garden of Forking Prompts: How Users Explore Narrative Space in Story Generation. In Proceedings of the Conference on Language Modeling (COLM), Oct 2026, San Francisco, CA, <https://openreview.net/forum?id=w6VGVCw4FF>.
- [C.2] Rohan Das, Advait Deshmukh, Alexandria Leto, Zohar Naaman, I-Ta Lee, and Maria Leonor Pacheco. 2026. A Structured Clustering Approach for Inducing Media Narratives. Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026), San Diego, CA, <https://aclanthology.org/2026.acl-long.1970/>.
- [C.3] Advait Deshmukh, Ashwin Umadi, Dananjay Srinivas, and Maria Leonor Pacheco. 2025. All Entities are Not Created Equal: Examining the Long Tail for Ultra-Fine Entity Typing. Proceedings of the 14th Joint Conference on Lexical and Computational Semantics (*SEM 2025), co-located with EMNLP 2025, <https://aclanthology.org/2025.starsem-1.15>.
- [S.1] Mreedul Gupta, Advait Deshmukh, Ashwin Umadi, Matt Pauk, and Maria Leonor Pacheco. 2026. Narrative-UFET: Narrative Generation for Ultra-Fine Entity Typing. arXiv preprint, [arXiv:2606.27598](https://arxiv.org/abs/2606.27598).

SELECTED RESEARCH PROJECTS

A Structured Clustering Approach for Inducing Media Narratives

Collaborator

Spring 2025 – Summer 2025

BLAST Lab, CU Boulder

- Implemented a modified DAPrompt (Xiang et al.) based relationship classifier to extract causally related event pairs across 4 datasets.
- Infused knowledge from teacher models (Llama3, Qwen3) to aid in domain adaptation.

Examining the Long Tail for Ultra-Fine Entity Typing

Independent study advised by Dr. Maria Pacheco

Fall 2024 – Spring 2025

BLAST Lab, CU Boulder

- Conducted rigorous empirical analysis to study limitations of predominantly popular language model-based approaches for ultra-fine entity typing.
- Developed proxies to predict entity frequencies based on pretrained language models and analyzed its effect on typing performance.

Improving LENS with Knowledge Distillation

Independent study advised by Dr. Tom Yeh

Spring 2024

BAIC Lab, CU Boulder

- Devised a selective caption filtering module on top of the approach provided by LENS (Berrios et al.) which enabled text only Large Language Models (LLMs) to "see".
- Leveraged LLMs and knowledge distillation techniques to train a compact RoBERTa-based classifier model to streamline the filtering process.

TEACHING EXPERIENCE

University of Colorado Boulder

Teaching Assistant, CSCI 1300: Intro to Computing

Fall 2024 – Spring 2025

Boulder, CO

- Assisted in the course by grading, holding recitations & office hours to support students.
- Collaborated with faculty to redesign course materials, improving student engagement.

University of Colorado Boulder

Course Manager, CSCI 1300: Intro to Computing

Fall 2023 – Spring 2024

Boulder, CO

- Managed operations for the course, serving over 900 students through office hours and online platform Ed.
- Collaborated with faculty to create and grade assignments, projects & exams.

PROFESSIONAL EXPERIENCE

Amazon

Applied Science Intern

Summer 2026

Bellevue, WA

- Interned with Supply Chain Optimization Technologies (SCOT) on the long-term planning & forecasting team.
- Improved evaluation for their agentic systems.

Societe Generale Global Solution Centre

Software Development Engineer

Jul 2021 – Jul 2023

Bangalore, India

- Developed a Retrieval Augmented Generation (RAG) based chat system to assist support teams in efficiently handling policy queries.
- Built an innovation portal, generating 120 ideas across 7 departments to foster cross-departmental collaboration.

Centre for Development of Advanced Computing (CDAC)

Research Intern

Jul 2020 – Dec 2020

Pune, India

- Engineered a scalable platform for High Performance Computing (HPC) job submission to optimize data processing.
- Developed a portal for seamless management of big data tasks, enhancing research efficiency.

VoiceQube

Summer Intern

May 2019 – Jul 2019

Bangalore, India

- Participated in summer internship program as part of the practice school program at BITS Pilani.
- Developed a conversational service to assist students in their GRE preparation.

SKILLS

Programming Languages: Python, JavaScript, TypeScript, C++, Java

Frameworks & Tools: React, Spring, FastAPI, Flask

Machine Learning Libraries: PyTorch, Transformers, NLTK, Pandas, Numpy, Langchain, Ollama

Cloud Technologies: Google Cloud, AWS, Microsoft Azure

DevOps & Version Control: Docker, Git, Kubernetes

AWARDS AND GRANTS

Conference Support Funds – \$1500

Computer Science Department, University of Colorado

Dec 2025

Boulder, USA

Travel Subsidy Recipient – \$750

*SEM 2025, co-located with EMNLP 2025

Nov 2025

Suzhou, China

Lloyd Botway Fellow – \$500

Computer Science Department, University of Colorado

April 2025

Boulder, USA

Outstanding Teaching Assistant – \$500

CSCI 1300, University of Colorado

March 2025

Boulder, USA

Winner - Poster Competition – \$500

Annual Research Expo, University of Colorado

Feb 2025

Boulder, USA

Grad App Aid Recipient – \$500

QueerInAI

Nov 2024

Boulder, USA

LEADERSHIP EXPERIENCE

Founder and President

VoiceQube Pilani

Sep 2019 – May 2021

- Founded and led VoiceQube BITS Pilani Chapter to develop Voice interface-based projects.
- Deployed BITS Buddy to handle queries about campus life, responding to ~4000 requests between January and March 2020.